UWE Bristol

Industrial Control
UFMF6W-20-2
Control Systems Engineering
UFMEUY-20-3

Lecture 5: Block Diagrams and Steady State Errors

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Today's Lecture

- Block diagrams to represent control systems
- Block diagram manipulation
- Example
- Steady State Errors

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Block Diagrams

- Block Diagrams provide a pictorial representation of a system
- Unidirectional operational block representing individual transfer functions
- Three basic elements:
  - Rectangles - operators
  - Lines - signals
  - Circles - additional or subtraction

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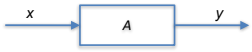
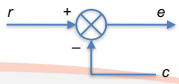
Block Diagrams: Examples

- $y = Ax$



- $e = r - c$



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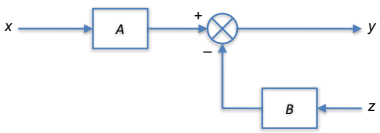
Block Diagrams: Examples

- $y = Ax-Bz$



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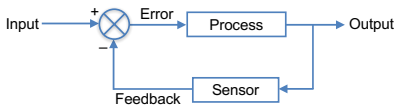
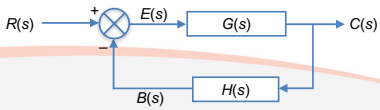
Closed Loop System

- Simple Closed Loop Control System





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Closed Loop System

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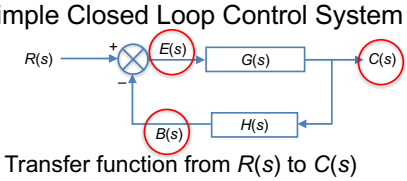

– Transfer function from $R(s)$ to $C(s)$

$$\left.\begin{array}{l} E(s)=R(s)-B(s) \\ B(s)=H(s)C(s) \\ C(s)=G(s)E(s)\rightarrow E(s)=\dfrac{C(s)}{G(s)} \end{array}\right\}\rightarrow \frac{C(s)}{G(s)}=R(s)-H(s)C(s)$$

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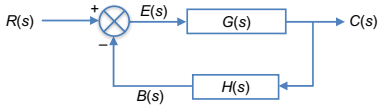
Closed Loop System

- Simple Closed Loop Control System



– Transfer function from $R(s)$ to $C(s)$

$$\frac{C(s)}{G(s)}=R(s)-H(s)C(s)\rightarrow C(s)\left(\frac{1}{G(s)}+H(s)\right)=R(s)$$

$$\boxed{\therefore \frac{C(s)}{R(s)}=\frac{G(s)}{1+G(s)H(s)}}$$

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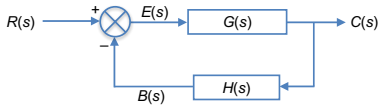
Closed Loop System

- Simple Closed Loop Control System



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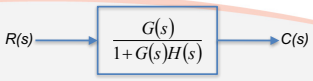

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Closed Loop System

- Simple Closed Loop Control System

– With unity feedback, $H(s)$ = 1

$$\frac{G(s)}{1+G(s)H(s)}\rightarrow\frac{G(s)}{1+G(s)}$$

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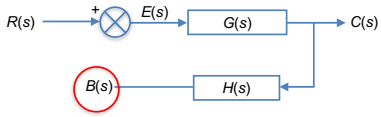
Open Loop Transfer Function

- Remove the feedback link from summing junction



$$E(s)=R(s)$$

Open Loop Transfer Function given by:

$$\frac{B(s)}{E(s)}=G(s)H(s)$$

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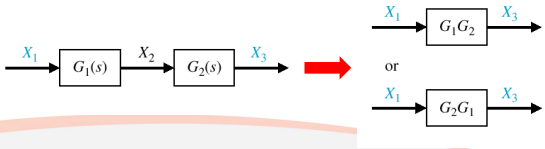
Block Diagram Manipulation

- Diagrams can be manipulated using the following transformations
- Combining Blocks in Series:



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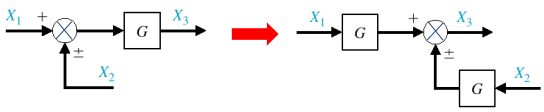
Block Diagram Manipulation

- Moving a summing junction



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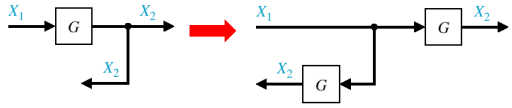
Block Diagram Manipulation

- Moving a pickoff point ahead of a block



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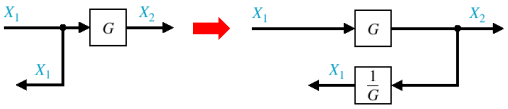
Block Diagram Manipulation

- Moving a pickoff point behind a block



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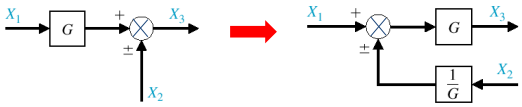
Block Diagram Manipulation

- Moving a summing point ahead of a block



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Block Diagram Manipulation

- Eliminating a feedback loop

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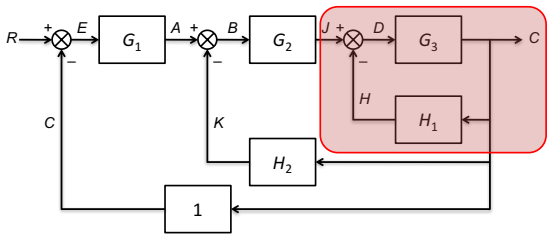
Example



Consider subgroup containing G_3 and H_1:

$$\frac{C}{J} = \frac{G_3}{1 + G_3 H_1} = G_4$$

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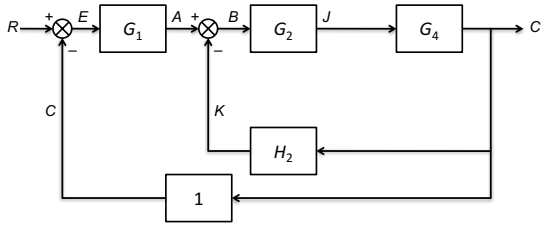
Example



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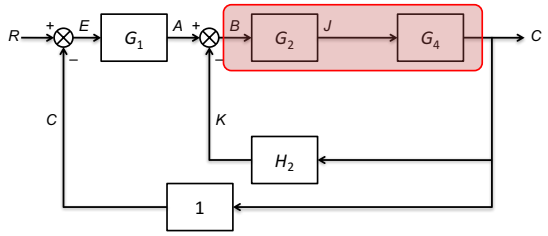
Example



Consider subgroup containing G_2 and G_4:

$$\frac{C}{B} = G_2G_4 = G_5$$

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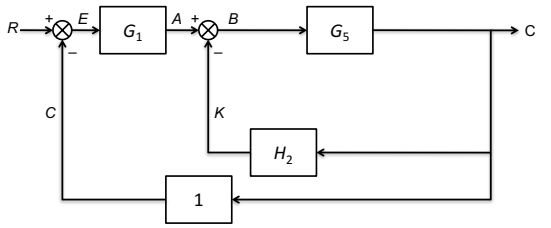
Example



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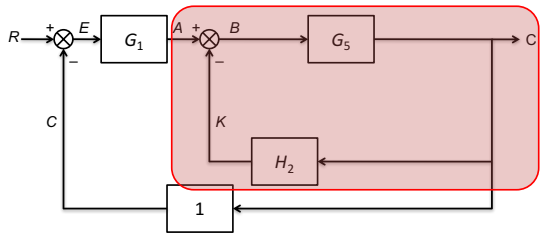
Example



Consider subgroup containing G_5 and H_2:

$$\frac{C}{A} = \frac{G_5}{1+G_5H_2} = G_6$$

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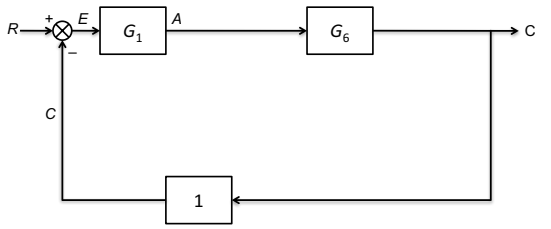
Example



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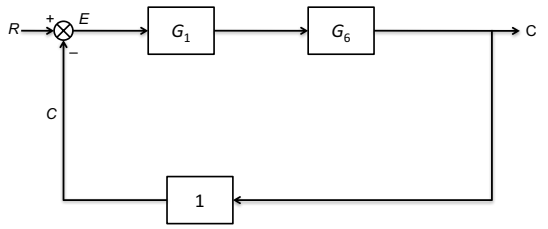
Example



Final Closed Loop Transfer Function

$$\frac{C}{R} = \frac{G_1G_6}{1+G_1G_6} = \frac{G_1G_2G_3}{1+G_3(H_1+G_2H_2)+G_1G_2G_3}$$

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Steady State Errors

- Feedback control used to reduce steady-state errors
- Steady-state error is error after the transient response has decayed
- If error is unacceptable, the control system will need modification
- Errors are evaluated using standardised inputs
  - Step inputs
  - Ramp inputs
  - Sinusoidal inputs

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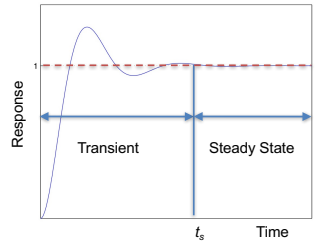
Example – No SS Error



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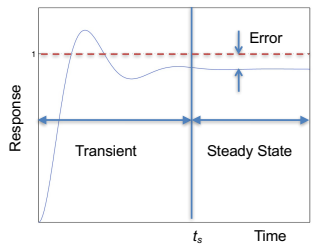
Example – No SS Error



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Causes of Steady State Error

- Errors can be caused by factors including
  1. Instrumentation of measurement errors
  2. System non-linearities – deadbands, hysteresis, saturation etc.
  3. Form of input signal
  4. Form of system transfer function
  5. External disturbances acting on the system, for example: forces or torques

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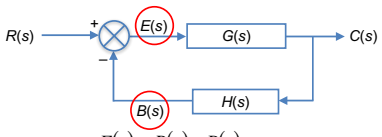
Error Function



$$E(s) = R(s) - B(s)$$

$$B(s) = H(s)C(s) = H(s)G(s)E(s)$$

$$E(s) = R(s) - G(s)H(s)E(s)$$

$$E(s)[1 + G(s)H(s)] = R(s)$$

$$\boxed{\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}}$$

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Calculating Value

System dependent

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)} \rightarrow E(s) = \frac{1}{1 + G(s)H(s)} \times R(s)$$

Input dependent

- Use the final value theorem:

$$E_{ss} = \lim_{t \to \infty} E(t) = \lim_{s \to 0} sE(s)$$

- Inputs can be
  - Step $\longrightarrow$ $R(s) = \frac{A}{s}$ ← A is step amplitude
  - Ramp $\longrightarrow$ $R(s) = \frac{A}{s^2}$ ← A is step velocity

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Example

- System and Feedback Transfer functions:

$$G(s)=\frac{1}{s(1+\tau s)} \quad \text{and} \quad H(s)=1$$

- Error

$$\frac{E(s)}{R(s)}=\frac{1}{1+G(s)H(s)}=\frac{1}{1+\frac{1}{s(1+\tau s)}}=\frac{s(1+\tau s)}{s(1+\tau s)+1}$$

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Example

- Step input

$$\frac{E(s)}{R(s)}=\frac{s(1+\tau s)}{s(1+\tau s)+1} \rightarrow E(s)=\frac{A}{s}\frac{s(1+\tau s)}{s(1+\tau s)+1}$$

- Steady State Error:

$$E_{ss}=\lim_{s\to 0} sE(s)=\cancel{s}\frac{A}{\cancel{s}}\frac{s(1+\tau s)}{s(1+\tau s)+1}=\frac{0(1+0\tau)A}{0(1+0\tau)+1}=0$$

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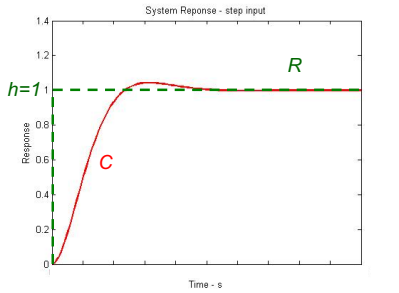
- as E_{ss} = 0 and $H(s)$ = 1, steady state output will be same as input



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Example

- Ramp input

$$\frac{E(s)}{R(s)}=\frac{s(1+\tau s)}{s(1+\tau s)+1} \rightarrow E(s)=\frac{A}{s^2}\frac{s(1+\tau s)}{s(1+\tau s)+1}$$

- Steady State Error:

$$E_{ss}=\lim_{s\to 0} sE(s)=\cancel{s}\frac{A}{\cancel{s^2}}\frac{\cancel{s}(1+\tau s)}{s(1+\tau s)+1}=\frac{(1+0\tau)A}{0(1+0\tau)+1}=A$$

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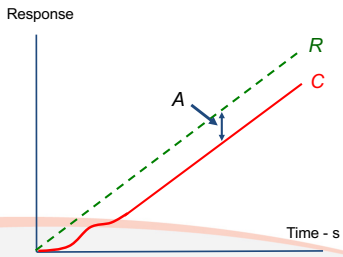
- in this case E_{ss} is not zero and during application of ramp input C will lag R by A



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Example

- System and Feedback Transfer functions:

$$G(s)=\frac{1}{1+\tau s} \quad \text{and} \quad H(s)=2$$

- Error

$$\frac{E(s)}{R(s)}=\frac{1}{1+G(s)H(s)}=\frac{1}{1+\frac{2}{1+\tau s}}=\frac{\tau s+1}{\tau s+3}$$

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Example

- Step input

$$\frac{E(s)}{R(s)} = \frac{\tau s+1}{\tau s+3} \rightarrow E(s) = \frac{A}{s}\frac{\tau s+1}{\tau s+3}$$

- Steady State Error:

$$E_{ss} = \lim_{s\to 0} sE(s) = \cancel{s}\frac{A}{\cancel{s}}\frac{\tau s+1}{\tau s+3} = \frac{(0\tau+1)A}{0\tau+3} = \frac{A}{3}$$

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- in this case, error is not zero and the output will not be equal to the input
- steady state output C_{ss} can also be found using the *Final Value Theorem* as follows :

$$E(s) = R(s) - H(s)C(s) \;\rightarrow\; C(s) = \frac{R(s)-E(s)}{H(s)} = \left(1 - \frac{1+\tau s}{3+\tau s}\right)\frac{R(s)}{2}$$

$$C_{ss} = \lim_{s\to 0} sC(s) = \lim_{s\to 0} s\left(1 - \frac{1+\tau s}{3+\tau s}\right)\frac{A}{2s} = \left(1 - \frac{1}{3}\right)\frac{A}{2} = \frac{A}{3}$$

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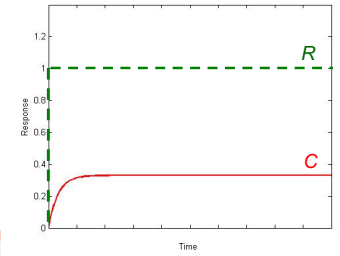
- resulting response for A = 1:



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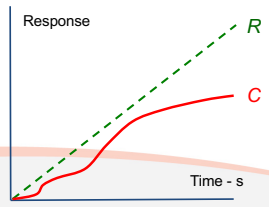
(b) *ramp input:*

$$E_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \cancel{s}\left(\frac{(1+\tau s)A}{(3+\tau s)s^{\cancel{2}}}\right) = \frac{(1+\tau\times 0)}{(3+\tau\times 0)}\frac{A}{0} = \infty$$

error continues to increase which is not acceptable



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Today's lecture

- Block Diagrams pictorial representation of control system
- Rectangles represent operations
- Lines are signals
- Summing junctions enable addition/subtraction
- Manipulation Techniques to reduce block diagrams to transfer function
- Steady State Errors help to determine what happens to signal in steady state

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Example Block Diagram

An electrical motor is used in a closed loop system to control the angular position of an inertial load. The position of the load, which is directly connected to the motor, is measured by a simple rotary potentiometer. The output signal from the transducer is compared with the input demand and the resulting error signal is passed to a voltage/current amplifier. The input demand is converted from angular displacement to voltage before being connected to the summing junction.

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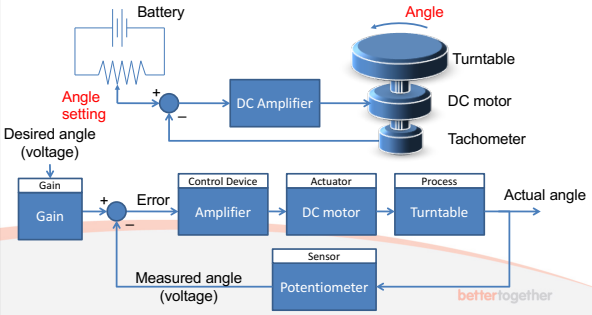
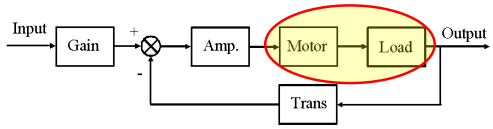
Example

- Closed Loop

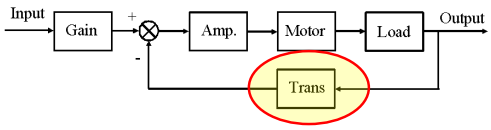


system block diagram:



An electrical motor is used in a closed loop system to control the angular position of an inertial load.

system block diagram:



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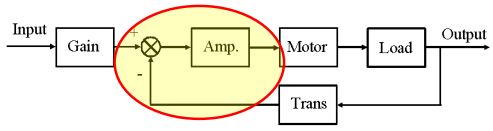
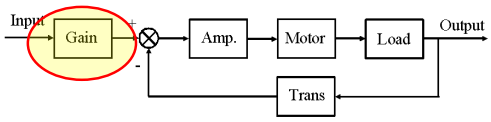
system block diagram:



The output signal from the transducer is compared with the input demand and the resulting error signal is passed to a voltage/current amplifier.

system block diagram:



The input demand is converted from angular displacement to voltage before being connected to the summing junction.

- system equations:

(a) Motor torque – $T_m = k_m I$

(b) Amplifier – $I = AV_e$

(c) Load – $T_L = Js^2\theta_o + cs\theta_o$

(d) Input demand – $V_i = k_p\theta_i$

(e) Feedback – $V_o = k_T\theta_o$

(f) Error – $V_e = V_i - V_o$

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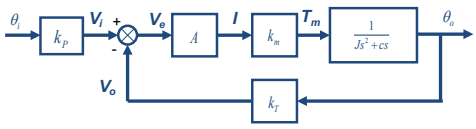
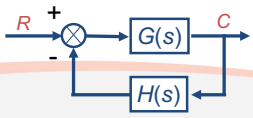
- system block diagram



reduced block diagram



$$G(s) = \frac{Ak_m k_P}{Js^2 + cs}$$

$$H(s) = \frac{k_T}{k_P}$$

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- *Closed Loop Transfer* function for system:

$$\frac{\theta_o}{\theta_i}(s) = \frac{Ak_m k_P}{Js^2 + cs + Ak_m k_T}$$

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